THE LOGARITHMIC FACTOR IN THE TUNED CONSTANT-STEP RATE OF RANDOM RESHUFFLING

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Abstract. For strongly convex finite sums, the known last-iterate upper bounds for random reshuffling with a constant step size chosen from the number of epochs carry a logarithmic factor, while the known lower bounds carry none. We show that for constant steps the factor cannot be removed. On a one-dimensional sum of n_1 quadratic and $n-n_1$ linear components, with average curvature μ, gradient bound G at the minimiser and initial distance D, every constant step $\gamma \le 1/L$ leaves $\mathbb{E}F(x_K) - F^* \ge G^2(\Lambda - \ln\Lambda)^2/(512\,\mu T^2)$ after K epochs, where $T = nK$ and $\Lambda = \ln(\mu TD/G)$, while the step $\gamma = \Lambda/(\mu T)$ gives at most $G^2\Lambda^2(1+o(1))/(8\mu T^2)$ once K is large compared with $n_1\Lambda$. The proof computes one epoch exactly: the stationary mean of the epoch-end iterate is $\gamma G\varphi/2$, where φ is a weighted mean lying between $(1-\gamma L)^{n_1-1}$ and 1, and the epoch variance reduces to the variance of a Mann–Whitney statistic. The bound concerns the worst starting point: from the other side of the minimiser a tuned step can cancel the transient against the bias, while a start of random sign keeps the bound for every step.

1. Introduction

Let $F = \frac{1}{n}\sum_{i=1}^n f_i$ be a finite sum with L-smooth components and a μ-strongly convex average, with minimiser x^*. Random reshuffling (RR) runs K epochs; each epoch draws a fresh uniform permutation π of $\{1,\dots,n\}$ and makes the n steps $x \leftarrow x - \gamma\nabla f_{\pi(t)}(x)$, $t = 1,\dots,n$. We write $T = nK$ for the total number of steps and x_K for the iterate at the end of epoch K.

Upper bounds for the last iterate, or for function values along the last epoch, with a constant step chosen from K, are given by Mishchenko, Khaled and Richtárik [6], Ahn, Yun and Sra [1] and Liu and Zhou [4], among others; each carries a logarithmic factor in K or T, coming from the step size. For convex smooth components, Chen and Zhang [2] obtain a last-iterate bound with constant component steps of order $1/(L(1+\log(nK)))$, and they too leave the removal of the logarithm open. Lower bounds for constant steps are due to Safran and Shamir [8, 9], Rajput, Gupta and Papailiopoulos [7] and Cha, Lee and Yun [3]; they carry no logarithm. Liu and Zhou write that the extra factor $1+\log K$ in their step sizes "seems necessary in our analysis" and ask whether it can be removed [4, Section 6].

The result below answers this for constant steps on a simple instance. The mechanism is the competition between the transient and the bias. After K epochs the mean iterate is $A^K x_0 + (1-A^K)m$, where $A \approx e^{-\gamma\mu n}$ is the epoch contraction and m is a stationary bias proportional to γ. Forgetting the start requires $\gamma\mu T \gtrsim \ln(\mu TD/G)$, and then the bias is of order $G\ln(\mu TD/G)/(\mu T)$. What has to be proved is that the bias has a definite sign and size for every step $\gamma \le 1/L$, uniformly in n, and that the fluctuations of the epoch do not change the order of the error at the tuned step. Both follow from an exact computation of one epoch.

The instance. Fix $n \ge 2$, $1 \le n_1 \le n/2$, $n_2 = n - n_1$ and $L, s > 0$. Let

$$f_i(x) = \frac{L}{2}x^2 - sx \quad (1 \le i \le n_1), \qquad f_i(x) = s\,\frac{n_1}{n_2}\,x \quad (n_1 < i \le n).$$

The linear parts cancel in the average, so $F(x) = \frac{\mu}{2}x^2$ with $\mu = Ln_1/n$, $x^* = 0$ and $F^* = 0$. Every component is convex and L-smooth. At the minimiser the gradients are $-s$ and sn_1/n_2, so $G :=$

$\max_i |f_i'(x^*)| = s$ and $\sigma_*^2 := \frac{1}{n}\sum_i f_i'(x^*)^2 = s^2 n_1/n_2$. The condition number is $\kappa = L/\mu = n/n_1$. We call the first n_1 components stiff and the others flat. For $x_0 = D > 0$ put

$$\Lambda = \ln \frac{\mu T D}{G}.$$

Theorem 1.1. *Let $K \geq 2$ and $\Lambda \geq 3$.*

(a) *If $K \geq n_1(\Lambda - \ln\Lambda)/2$, then every constant step $0 < \gamma \leq 1/L$ gives*

$$\mathbb{E}F(x_K) - F^* \;\geq\; \frac{G^2(\Lambda - \ln\Lambda)^2}{512\,\mu T^2}.$$

(b) *If $K \geq 2\Lambda$, the step $\gamma = \Lambda/(\mu T)$ gives*

$$\mathbb{E}F(x_K) - F^* \;\leq\; \frac{G^2}{2\mu T^2}\Big[\Big(1 + \frac{\Lambda}{2}\Big)^2 + \frac{\Lambda^3 n_1}{K}\Big(\frac{1}{2} + \frac{\Lambda\sqrt{n_1}}{K}\Big)^2\Big].$$

In particular, if $K \geq \max(2\Lambda, n_1\Lambda)$, then $\min_{0<\gamma\leq 1/L} \mathbb{E}F(x_K) - F^* = \Theta\big(G^2 \ln^2(\mu T D/G)/(\mu T^2)\big)$ with absolute constants.

The lower bound is in terms of G. With $n_1 = 1$ it reads in terms of σ_*: then $G^2 = (n-1)\sigma_*^2$, $\mu = L/n$ and $\Lambda = \ln(LKD/G)$, so

$$\min_{0<\gamma\leq 1/L} \mathbb{E}F(x_K) - F^* \;\geq\; \frac{\sigma_*^2\,(\Lambda - \ln\Lambda)^2}{1024\,\mu n K^2},$$

which has the dependence on n and K of the constant-step lower bound $L\sigma_*^2/(\mu^2 n K^2)$ of [9, 3], multiplied by $\ln^2 K$; the factor $\kappa = L/\mu$ of that bound does not appear. The matching upper bound of Theorem 1.1(b) shows that on this family no power of κ appears; whether the logarithm and a positive power of κ can occur together in a lower bound for the tuned last iterate is not addressed here.

Section 2 computes one epoch, Section 3 proves part (a) together with a version for a start of random sign, Section 4 proves part (b), and Section 5 reports exact computations of $\mathbb{E}F(x_K) - F^*$ that check the constants.

2. One epoch

Write $r = 1 - \gamma L$. A stiff step maps x to $rx + \gamma s$ and a flat step maps x to $x - \gamma s n_1/n_2$. In an epoch with permutation π, let q_t be the number of stiff steps that come after step t, and let $c_t = \gamma s$ if step t is stiff and $c_t = -\gamma s n_1/n_2$ if it is flat. Composing the n maps gives

$$X_{k+1} = AX_k + B_k, \qquad A = r^{n_1}, \qquad B_k = \sum_{t=1}^{n} c_t r^{q_t}, \tag{1}$$

where X_k is the iterate at the end of epoch k. The coefficient A is deterministic, and the B_k are independent and identically distributed, each independent of X_k, because each epoch draws a fresh permutation. Hence

$$\mathbb{E}X_K = A^K X_0 + (1 - A^K)\,m, \qquad m = \frac{\mathbb{E}B}{1-A}, \qquad \operatorname{Var} X_K = \frac{1 - A^{2K}}{1 - A^2}\operatorname{Var} B \tag{2}$$

for a deterministic X_0 and $0 \leq r < 1$, and $\mathbb{E}F(x_K) - F^* = \frac{\mu}{2}\mathbb{E}X_K^2$.

Lemma 2.1. *For $0 \leq r < 1$,*

$$m = \frac{\gamma s}{2}\,\varphi(r), \qquad \varphi(r) = \frac{2}{n_1 + 1} \cdot \frac{\sum_{k=0}^{n_1-1}(k+1)r^k}{\sum_{k=0}^{n_1-1} r^k},$$

and

$$\max\Big(r^{n_1-1}, \frac{2}{n_1+1}\Big) \;\leq\; \varphi(r) \;\leq\; 1.$$

Proof. In a uniform permutation a stiff step has q uniform on $\{0,\dots,n_1-1\}$, and a flat step, which falls into one of the n_1+1 gaps between stiff steps with equal probability, has q uniform on $\{0,\dots,n_1\}$. With $S_j=\sum_{q=0}^{j-1} r^q$,

$$\mathbb{E}B=\gamma s\,S_{n_1}-\gamma s\,\frac{n_1}{n_1+1}\,S_{n_1+1}=\frac{\gamma s}{n_1+1}\sum_{q=0}^{n_1-1}\bigl(r^q-r^{n_1}\bigr).$$

Since $r^q-r^{n_1}=(1-r)\sum_{p=0}^{n_1-q-1} r^{q+p}$ and the number of pairs (q,p) with $q+p=k\le n_1-1$ is $k+1$,

$$\sum_{q=0}^{n_1-1}(r^q-r^{n_1})=(1-r)\sum_{k=0}^{n_1-1}(k+1)r^k,$$

while $1-A=1-r^{n_1}=(1-r)\sum_{k=0}^{n_1-1} r^k$. Dividing gives the formula for m.

The quantity $\varphi(r)$ is the mean of the increasing sequence $a_k=2(k+1)/(n_1+1)$, $k=0,\dots,n_1-1$, under the non-increasing weights r^k. By Chebyshev's sum inequality it is at most the unweighted mean of the a_k, which is 1. It is at least the smallest a_k, which is $2/(n_1+1)$. Finally $\sum_k(k+1)r^k\ge r^{n_1-1}n_1(n_1+1)/2$ and $\sum_k r^k\le n_1$ give $\varphi(r)\ge r^{n_1-1}$. □

At $r=1$ the formula gives $\varphi=1$, so $m\to\gamma s/2$ as $\gamma\to 0$; the bias is positive, and it is never larger than this limit.

Lemma 2.2. *If $0\le\gamma L\le 1$, then*

$$\operatorname{Var}B\;\le\;\gamma^4L^2s^2\Bigl[\Bigl(\frac{n_1^3(n+1)}{12\,n_2}\Bigr)^{1/2}+\gamma L n_1^3\Bigr]^2.$$

Proof. Since $\sum_t c_t=n_1\gamma s-n_2\gamma s n_1/n_2=0$, we may write $B=\sum_t c_t(r^{q_t}-1)$. For $x=\gamma L\in[0,1]$ and an integer $q\ge 0$, Bernoulli's inequality and $e^{-y}\le 1-y+y^2/2$ ($y\ge 0$) give

$$0\;\le\;e_q:=r^q-1+q\gamma L\;\le\;e^{-q\gamma L}-1+q\gamma L\;\le\;\tfrac12(q\gamma L)^2\;\le\;\tfrac12(n_1\gamma L)^2.$$

Hence $B=E_1+E_2$ with $E_1=-\gamma L\sum_t c_tq_t$ and $E_2=\sum_t c_te_{q_t}$, and $|E_2|\le\frac12(n_1\gamma L)^2\sum_t|c_t|=\gamma^3sL^2n_1^3$, because $\sum_t|c_t|=2\gamma sn_1$.

In E_1 the stiff steps take the values $q=0,1,\dots,n_1-1$ once each, so their part is the constant $\gamma s\,n_1(n_1-1)/2$. For the flat steps, $\sum_{t\ \mathrm{flat}}q_t=U$ is the number of pairs (flat step, stiff step) in which the stiff step comes later. This is the Mann–Whitney statistic of two samples of sizes n_2 and n_1 under a uniform random order [5], whose variance is $n_1n_2(n+1)/12$. Therefore

$$\operatorname{Var}E_1=\gamma^2L^2\Bigl(\frac{\gamma sn_1}{n_2}\Bigr)^2\frac{n_1n_2(n+1)}{12}=\gamma^4L^2s^2\,\frac{n_1^3(n+1)}{12\,n_2},$$

and the claim follows from $\operatorname{Var}(E_1+E_2)^{1/2}\le\operatorname{Var}(E_1)^{1/2}+\sup|E_2|$. □

The first-order term is sharp: the exact epoch variance, computed in Section 5, is 0.91 to 1.00 times $\gamma^4L^2s^2n_1^3(n+1)/(12n_2)$ for $\gamma Ln_1\le 0.1$.

3. The lower bound

Proof of Theorem 1.1(a). Since $\mathbb{E}X_K^2\ge(\mathbb{E}X_K)^2$, it suffices to bound $\mathbb{E}X_K$ from below. For $0<\gamma\le 1/L$ we have $A\in[0,1)$ and, by Lemma 2.1, $m>0$; with $X_0=D>0$ both terms of $\mathbb{E}X_K=A^KD+(1-A^K)m$ are nonnegative, so $\mathbb{E}X_K$ is at least each of them. Put $u=\gamma\mu T=\gamma Ln_1K$. We show $\mathbb{E}X_K\ge G(\Lambda-\ln\Lambda)/(16\mu T)$, which gives the claim, since $\frac{\mu}{2}\bigl(G(\Lambda-\ln\Lambda)/(16\mu T)\bigr)^2=G^2(\Lambda-\ln\Lambda)^2/(512\mu T^2)$.

Steps with $\gamma L n_1 \le 1/2$. Then $\gamma L \le 1/2$, and $\ln(1-x) \ge -2x$ on $[0,1/2]$ gives $A^K = (1-\gamma L)^{n_1 K} \ge e^{-2u}$. Also $\varphi(r) \ge r^{n_1-1} \ge 1-(n_1-1)\gamma L \ge 1/2$, so $m \ge \gamma s/4$. If $u \le (\Lambda - \ln\Lambda)/2$, then

$$\mathbb{E}X_K \;\ge\; De^{-2u} \;\ge\; De^{-\Lambda+\ln\Lambda} = \frac{G\Lambda}{\mu T}.$$

If $u > (\Lambda-\ln\Lambda)/2$, then $u > \ln 2$, because $\Lambda - \ln\Lambda \ge 3 - \ln 3 > 2\ln 2$; hence $A^K \le e^{-\gamma L n_1 K} = e^{-u} < 1/2$ and

$$\mathbb{E}X_K \;\ge\; \tfrac{1}{2}m \;\ge\; \frac{\gamma s}{8} = \frac{uG}{8\mu T} \;>\; \frac{G(\Lambda-\ln\Lambda)}{16\,\mu T}.$$

Steps with $\gamma L n_1 > 1/2$. Then $A^K \le e^{-\gamma L n_1 K} \le e^{-K/2} < 1/2$ because $K \ge 2$, and $\varphi \ge 2/(n_1+1)$ gives $m \ge \gamma s/(n_1+1)$. Hence

$$\mathbb{E}X_K \;\ge\; \frac{\gamma s}{2(n_1+1)} \;>\; \frac{s}{4Ln_1(n_1+1)} \;\ge\; \frac{G}{8Ln_1^2} = \frac{G}{8\mu n n_1}.$$

Since $T = nK$, this is at least $G(\Lambda-\ln\Lambda)/(16\mu T)$ exactly when $K \ge n_1(\Lambda-\ln\Lambda)/2$. □

The start $x_0 = D$ lies on the side of the minimiser towards which the bias points, so the transient and the bias add. From $x_0 = -D$ they have opposite signs and a step can make $\mathbb{E}X_K$ vanish; the error is then carried by the variance alone, and the best step does much better (Section 5). The bound is therefore a statement about the worst start. It does not depend on choosing the start after the step, as the following version shows.

Proposition 3.1. *Let $x_0 = \pm D$ with probability $1/2$ each, independent of the permutations, and let K and Λ be as in Theorem 1.1(a). Then every constant step $0 < \gamma \le 1/L$ gives $\mathbb{E}F(x_K) - F^* \ge G^2(\Lambda-\ln\Lambda)^2/(512\,\mu T^2)$.*

Proof. By (1), $X_K = A^K X_0 + Z$ with $Z = \sum_{k<K} A^{K-1-k}B_k$ independent of X_0 and $\mathbb{E}Z = (1-A^K)m$. Since $\mathbb{E}X_0 = 0$,

$$\mathbb{E}X_K^2 = A^{2K}D^2 + \mathbb{E}Z^2 \;\ge\; \max\big((A^K D)^2,\; ((1-A^K)m)^2\big).$$

The proof of Theorem 1.1(a) bounded $\mathbb{E}X_K$ below by one of $A^K D$ and $(1-A^K)m$ in each case, so the same lower bound holds for $\mathbb{E}X_K^2$. □

4. The upper bound

Proof of Theorem 1.1(b). Take $\gamma = \Lambda/(\mu T)$, so $\gamma L n_1 = \Lambda/K \le 1/2$ and $0 < r < 1$. By (2), Lemma 2.1 and $A^K \le e^{-\gamma L n_1 K} = e^{-\Lambda} = G/(\mu T D)$,

$$0 \;\le\; \mathbb{E}X_K \;\le\; A^K D + m \;\le\; \frac{G}{\mu T} + \frac{\gamma s}{2} = \frac{G}{\mu T}\Big(1 + \frac{\Lambda}{2}\Big).$$

For the variance, $1 - A^2 \ge 1 - e^{-2\gamma L n_1} \ge \gamma L n_1$ because $1 - e^{-y} \ge y/2$ on $[0,1]$. Since $n_2 \ge n/2$ and $n \ge 2$, we have $(n+1)/n_2 \le 3$, so the first term in the bracket of Lemma 2.2 is at most $n_1^{3/2}/2$, and the second is $\gamma L n_1^3 = n_1^{3/2}\cdot \Lambda\sqrt{n_1}/K$. Hence

$$\operatorname{Var} X_K \;\le\; \frac{\operatorname{Var} B}{1-A^2} \;\le\; \gamma^3 L s^2 n_1^2 \Big(\frac{1}{2} + \frac{\Lambda\sqrt{n_1}}{K}\Big)^2 = \frac{s^2\Lambda^3 n n_1}{\mu^2 T^3}\Big(\frac{1}{2} + \frac{\Lambda\sqrt{n_1}}{K}\Big)^2,$$

using $L n_1^2 = \mu n n_1$. Adding $\frac{\mu}{2}(\mathbb{E}X_K)^2$ and $\frac{\mu}{2}\operatorname{Var} X_K$ and using $n/T = 1/K$ gives the bound. □

Proof of the last statement of Theorem 1.1. If $K \ge \max(2\Lambda, n_1\Lambda)$, both parts apply, since $n_1(\Lambda - \ln\Lambda)/2 \le n_1\Lambda$. In part (b), $\Lambda\sqrt{n_1}/K \le 1/\sqrt{n_1} \le 1$ and $\Lambda^3 n_1/K \le \Lambda^2$, so the bracket is at most $(1+\Lambda/2)^2 + \frac{9}{4}\Lambda^2 \le 4\Lambda^2$ for $\Lambda \ge 3$. In part (a), $\Lambda - \ln\Lambda \ge \Lambda/2$ for $\Lambda \ge 3$. Hence the minimum lies between $G^2\Lambda^2/(2048\mu T^2)$ and $2G^2\Lambda^2/(\mu T^2)$. □

Remark 4.1. Theorem 1.1 covers steps $0 < \gamma \le 1/L$. For $1/L < \gamma < 2/L$ the factor r is negative, the weights in φ alternate in sign, and Lemma 2.1 does not apply; these steps are not treated here. The analyses cited above work with $\gamma \le 1/L$ or smaller.

Remark 4.2. The theorem concerns constant steps. A step schedule decreasing within the run is not covered, and the logarithm in Theorem 1.1 is a property of the best constant step, not of random reshuffling as such.

5. Numerical checks

All quantities below are exact up to rounding. For each step γ the moments $\mathbb{E}B$ and $\mathbb{E}B^2$ of one epoch are computed by a dynamic programme over the position in the epoch and the number of stiff steps already taken, and $\mathbb{E}X_K$, $\mathbb{E}X_K^2$ follow from (1) by recursion over the K independent epochs. The programme agrees with direct simulation of random reshuffling with $4 \cdot 10^5$ runs to within the sampling error. We use $L = s = D = 1$.

The epoch. Over $n_1 \le 200$ and a grid of $r \in [0,1]$, $\varphi(r) \le 1$ and $\varphi(r) \ge \max(r^{n_1-1}, 2/(n_1+1))$; for $n_1 = 1$ both bounds are equalities. The stationary mean of the programme equals $(\gamma s/2)\varphi(r)$ to twelve digits. The exact $\operatorname{Var} B$ lies below the bound of Lemma 2.2 in every case tried and is 0.91 to 1.00 times its first-order term.

The rate. For $(n, n_1) \in \{(40,1), (100,10), (100,50), (1000,100)\}$ and $K \in \{10^3, 10^4\}$, the minimum over a grid of steps in $(0, 1/L]$ of the exact error from $x_0 = D$ is 1.29 to 1.54 times $G^2\Lambda^2/(8\mu T^2)$, the leading term of part (b), and the ratio decreases as K grows (1.54 to 1.40 for $(40,1)$, 1.42 to 1.32 for $(100,10)$). The minimum is attained at $\gamma\mu T/\Lambda$ between 1.04 and 1.10, and at $\gamma = \Lambda/(\mu T)$ the error lies below the bound of part (b) in all six cases. The minimum exceeds the lower bound of part (a) by a factor of 126 to 190, so the constant $1/512$ is far from sharp. From $x_0 = -D$ the minimum is $2 \cdot 10^{-4}$ to 0.018 times the same unit, which is the cancellation described after the proof of Theorem 1.1(a).

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

Anthropic Claude AI systems were used to develop the mathematical derivations, write and run the numerical checks, perform internal consistency and literature reviews, and write and prepare this manuscript and its publication files. The author selected the problem and directed the work; the author did not carry out the mathematics or the writing. Responsibility for submission

or publication rests with the author. The AI-assisted internal reviews, compilation checks and rendered-page inspections are not external peer review or formal verification.

Data and code availability

The paper is theoretical. The script that produces the numerical checks of Section 5 is in the `code` directory of the repository

`github.com/yspennstate/rr-constant-step-log-factor`.

The paper is part of the Math Brain project, whose public repository is https://github.com/yspennstate/math-brain-public [10].

References

[1] K. Ahn, C. Yun and S. Sra, *SGD with shuffling: optimal rates without component convexity and large epoch requirements*, Advances in Neural Information Processing Systems 33 (NeurIPS 2020), arXiv:2006.06946.

[2] Z. Chen and Z. Zhang, *Large-step last-iterate convergence of random-reshuffling stochastic gradient descent*, arXiv:2610.19444 (2026).

[3] J. Cha, J. Lee and C. Yun, *Tighter lower bounds for shuffling SGD: random permutations and beyond*, Proc. 40th Int. Conf. Machine Learning (ICML 2023), arXiv:2303.07160.

[4] Z. Liu and Z. Zhou, *On the last-iterate convergence of shuffling gradient methods*, Proc. 41st Int. Conf. Machine Learning (ICML 2024), arXiv:2403.07723.

[5] H. B. Mann and D. R. Whitney, *On a test of whether one of two random variables is stochastically larger than the other*, Ann. Math. Statist. **18** (1947), 50–60.

[6] K. Mishchenko, A. Khaled and P. Richtárik, *Random reshuffling: simple analysis with vast improvements*, Advances in Neural Information Processing Systems 33 (NeurIPS 2020), arXiv:2006.05988.

[7] S. Rajput, A. Gupta and D. Papailiopoulos, *Closing the convergence gap of SGD without replacement*, Proc. 37th Int. Conf. Machine Learning (ICML 2020).

[8] I. Safran and O. Shamir, *How good is SGD with random shuffling?*, Proc. 33rd Conf. Learning Theory (COLT 2020), arXiv:1908.00045.

[9] I. Safran and O. Shamir, *Random shuffling beats SGD only after many epochs on ill-conditioned problems*, Advances in Neural Information Processing Systems 34 (NeurIPS 2021).

[10] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, https://github.com/yspennstate/math-brain-public.

[11] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24.

[12] Y. Shmalo, *Brain Framework*, public software repository, 2026, https://github.com/yspennstate/brain-framework-public.